3rd AGN School

Python for Astronomy and Astrophysics

PROBLEM SHEET & QUESTION BANK

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Q&A Community: Official Slack Channel

Course Guidelines & Collaboration

Welcome to the problem sheet for the **3rd AGN School**! I've put together these questions to help you get the hang of using Python for everyday astronomy tasks from basic math to calculating things like redshifts and black hole masses.

A few tips for your code:

- ✓ Try to keep your code clean and easy to read.
- ✓ Add comments so it's clear what your logic is doing.
- ✓ Use functions where it makes sense rather than writing one huge script.
- ✓ Remember to `return` values from your functions instead of just `print()`ing them.

Doubts & Submissions

Getting stuck is totally normal. If you hit a roadblock or have any questions, just drop your doubts in the **Slack channel** and we can discuss them there.

There's no strict deadline, so take your time. You can **submit your assignment at your leisure to my email** once you're happy with it.

1 Basic Problems

This section focus on basic variables, user inputs, lists, simple indexing, and string operations.

Q1. Write a program that stores the name, mass, and radius of the Sun in descriptive variables and prints them using Python f-strings.

Q2. Take a stellar temperature as user input and determine whether the star is hotter or cooler than our Sun ($T_\odot \approx 5778$ K).

Q3. Create a Python list containing the names of the planets of our Solar System and print them out sequentially using a `for` loop.

Q4. Store the names of five bright stars in a list. Print the first, the middle (third), and the last star using direct list indexing.

Q5. Given the string:

```
"Messier87"
```

use string slicing to programmatically extract:

- `Messier`
- `87`
- the fully reversed string (`78reisseM`)

2 Functions in Astronomy

Apply modular programming concepts by writing functional units to calculate key astronomical parameters.

Q6. Write a function:

```
redshift(lambda_obs, lambda_emit)
```

that returns the spectroscopic redshift z using the formula:

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{emit}}}{\lambda_{\text{emit}}}$$

Calculate the redshift of a distant galaxy whose Hα spectral line is observed at 700 nm instead of its rest-frame wavelength of 656.3 nm.

Q7. Write a function that converts a spectroscopic redshift z to a recession velocity v using the linear relationship:

$$v = cz$$

(Note: This approximation is valid for low-redshift objects, where speed of light $c \approx 3 \times 10^5$ km s^{-1}).

Q8. Write a function that calculates the luminosity distance d using Hubble's law:

$$d = \frac{v}{H_0}$$

Assume a Hubble constant value of:

$$H_0 = 70 \text{ km s}^{-1}\text{Mpc}^{-1}$$

Q9. Write a function that calculates the physical Schwarzschild radius (R_s) of a spherically symmetric body:

$$R_s = \frac{2GM}{c^2}$$

where G is the gravitational constant, M is the object's mass, and c is the speed of light.

Q10. Using the function written in the previous question, calculate and print the Schwarzschild radius for the following celestial bodies:

- The Earth ($M_\oplus \approx 5.97 \times 10^{24}$ kg)
- The Sun ($M_\odot \approx 1.989 \times 10^{30}$ kg)
- A supermassive black hole with a mass of $10^8 M_\odot$

Q11. Write a function that calculates the escape velocity v_e of a body:

$$v_e = \sqrt{\frac{2GM}{R}}$$

Q12. Calculate the escape velocities for all the planets in our Solar System and programmatically store the calculated results in a standard Python list.

3 Loops and Iterative Calculations in Astrophysics

In these exercises, you will handle physical equations across multiple stellar systems and coordinate configurations.

Q13. Given a Python list of stellar masses:

`[0.5, 1, 2, 5, 10, 20]` (in units of $M_\odot$)

calculate their corresponding luminosities using the empirical mass-luminosity relationship:

$$L \propto M^{3.5}$$

Print the results formatted neatly as a two-column table.

Q14. Use a loop to generate a sequence of stellar temperatures from 3000 K to 10000 K in steps of 500 K. For each temperature, compute the Wien's peak wavelength:

$$\lambda_{\max} = \frac{2.897 \times 10^{-3}}{T}\ \text{m}$$

Q15. Generate a table of black hole masses ranging from $10^5 M_\odot$ to $10^9 M_\odot$ (increasing in steps of $10^5 M_\odot$ or logarithmically) and compute their respective Schwarzschild radii.

Q16. Use nested loops to compute the individual pairwise gravitational forces between every possible pair of stars within a simulated small stellar cluster.

4 Intermediate Problems

These problems combine physical formulas with computational decision-making structures.

Q17. Write a modular function to compute stellar luminosity L based on its radius R and effective surface temperature T:

$$L = 4\pi R^2 \sigma T^4$$

where σ is the Stefan-Boltzmann constant.

Q18. Create a list containing arbitrary stellar temperatures. Using loops and conditional blocks (`if-elif-else`), automatically classify each star into its correct Morgan-Keenan spectral type (O, B, A, F, G, K, or M) based on standard physical temperature boundaries.

Q19. Write a function for the Eddington luminosity limit L_E:

$$L_E = 1.26 \times 10^{38} \left(\frac{M}{M_\odot}\right)\ \text{erg s}^{-1}$$

This is the theoretical limit at which the outward radiation pressure of a star/AGN balances the inward gravitational pull.

Q20. Given a list of observed active galactic nuclei (AGN) supermassive black hole masses, calculate and print their respective Eddington luminosities.

Q21. Write a function that calculates the distance modulus:

$$m - M = 5\log_{10}\left(\frac{d}{10}\right)$$

where m is the apparent magnitude, M is the absolute magnitude, and d is the distance to the source in parsecs. Use this function to compute distances to a set of stars with known astronomical magnitudes.

5 Advanced Challenges

These problems scale up complexity, calling for larger data processing structures and simulation design.

Q22. Given two aligned arrays representing the observed and emitted spectral wavelengths for 50 distant galaxies, write a program to calculate all 50 individual redshifts, and then find the overall average redshift of the sample.

Q23. Create a function that computes the characteristic age of the Universe (Hubble time t_H) based on the Hubble constant H_0:

$$t_H = \frac{1}{H_0}$$

Perform the dimensional analysis in your code and convert the final value into years.

Q24. Write a program that computes the orbital periods of planets/binary systems using Kepler's Third Law:

$$P^2 = \frac{4\pi^2 a^3}{GM}$$

where P is the orbital period, a is the semi-major axis, and M is the total mass.

Q25. Using nested loops, generate a complete distance matrix showing the mutual Euclidean distances between every possible pair of galaxies in a model galaxy catalog coordinate system.

Q26. Mini Project: Interactive AGN Catalogue Builder
Design a comprehensive Python program that compiles a database of Active Galactic Nuclei (AGN). The catalog must track:

- Object Name
- Spectroscopic Redshift (z)
- Supermassive Black Hole Mass ($M_\bullet$)

Store the items inside structural arrays or dictionaries. Develop custom, high-order functions to automatically compute and populate all derived physical properties (such as recession velocity, distance, and Eddington luminosity) for every record in your database.